

AN ASSESSMENT OF USAGE BEHAVIOURAL PATTERN OF UZHAVAN APP USERS (FARMERS) IN TAMIL NADU

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ABSTRACT

Before two decades information technology and agriculture were considered incompatible but today in this era of globalization. Mobile applications have become familiar among the people but in the field of agriculture, m-apps were in the stage of budding. As a step to deliver quality, timely and accurate agricultural information, Tamil Nadu Government has launched bi-lingual (Tamil and English) Uzhavan mobile application. *Ex-post facto* research design was adopted for the study and primary data was collected from a sample of 90 uzhavan app users in Thanjavur district of Tamil Nadu, India during January 2020 by personal interview with the help of well-structured and pre-tested ¹interview schedule prepared for the study. The result showed that uzhavan app users had browsed utmost seven sections and spent 6-10 minutes per day to browse the contents in the uzhavan app. Subsidy section, benefit registration and crop insurance had been the most often sought subjects by the uzhavan app users. Uzhavan app has relatively lowered the information searching time and it has created more awareness about agricultural government schemes.

KEYWORDS: Uzhavan app, m-agri app, farmers, ICT, e-agriculture and utility behavior.

INTRODUCTION

ICTs have become the essential factor for agricultural and educational development in the current century (Brown, 2002; Kozma and Anderson, 2002; Goodison, 2003; Kangro and Kangro, 2004; Hennessy *et al.*, 2005; Kennedy *et al.*, 2006; Bayindir and Inan, 2009) which is a powerful tool in addressing agricultural needs (Tshabalala, 2001). Twenty-first century would be deliberated as the age of information

and communication technology (ICT). According to (Zahedi, 2012) ICT was an extensive term that comprise computer hardware and software, digital broadcast and telecommunication technologies as well as digital information repositories online or offline and also includes social networking aspects on the web besides online file sharing system (Allahyari, 2010). A smartphone is an e-toy designed for the lonely inner child hidden in each and every one of us.

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Smart-phones were extensively considered as a prospective tool for communication and disseminating information. Ganesan (2013) stated that now-a-days mobile applications have become familiar among the people but in the field of agriculture m-apps were in the stage of budding. Aravindh Kumar S and Karthikeyan(2019 and 2020) stated that education and income levels are strong determinants, not only of whether or not people use the Internet, but also of how they use it. Internet users with higher levels of education use more advanced services, such as e-commerce and online financial and government services, to a higher degree than Internet users with lower levels of education and income levels, who use the Internet predominantly for communication and entertainment purposes. Social media is a platform of enlargement, and for agricultural producers the major reason for producing such a platform is mass influence (Varner, 2012). It gives farmers direct opportunity to connect with customers(Carr, 2015) and the key values of communication provided by social media as peer-to-peer networking (Stanley, 2013). The fast penetration of ITC thus brings new opportunities for farmers to improve their knowledge and livelihoods (Asongu, 2015)&(Aker, 2010). Farmers conflicting perceptions (Fuess, 2011), lack of capacity (Newbury, 2014) in using social media act as the constraint to use social media in extension programme deliver, Farmers' neither technology savvy nor receiving quality information (FAO, 2019), thus leading to reluctance in the process of adoption of digital tools' (Diem, 2011). As a step to deliver quality, timely and accurate agricultural information, Tamil Nadu Government has launched bi-lingual (Tamil and English) Uzhavan mobile application

in a bid to use technology for benefit of farmers. It was launched by Chief Minister of Tamil Nadu in the state capital Chennai on April 8, 2018. The size of the Uzhavan app was just 3.93 mb and it was made freely available to the people in Google play store and Apple store. Nearly 55, 55,587 people had downloaded the Uzhavan application. (ARISNET, 2020)²Uzhavan app a great initiative taken by Tamil Nadu government to deliver agricultural information, create awareness about agricultural schemes, disseminate market prices, weather forecast and also to establish contact between farmers and extension officials quicker than before. The objective of the study is to establish the usage pattern of uzhavan app. by the uzhavan app users (farmers).

MATERIAL AND METHODOS

An ex-post facto research design was adopted for the study. Sampling was done during December 2019, there were 38 districts in Tamil Nadu and Villupuram district counted more number of Uzhavan app users (AGRISNET, 2020). ³Villupuram district was bifurcated into Villupuram and Kallakurichi districts during November 2019. However the Uzhavan app users was not available for the bifurcated Villupuram district. Hence, Thanjavur district which ranked second in terms of the number of Uzhavan app users in the state was purposively selected. Thanjavur district had 14 blocks among them, two blocks namely, Patukottai and Peravurani were selected for the study as it had relatively more number of Uzhavan app users. The list of Uzhavan app users in the two selected blocks was obtained from the Assistant Director of Agriculture, Thanjavur. From the list, a total of 90 Uzhavan app users comprising 60 from Patukottai block and 30 from Peravurani block

were selected by using proportionate random sampling method. Data was collected from the selected Uzhavan app users by personal interview with the help of well-structured and pre-tested interview schedule. After gathering the required information, data were statistically analyzed using the SPSS version 16.0.

RESULT AND DISCUSSION

Usage pattern of uzhavan app by the uzhavan app users (farmers)

Usage pattern in the sense of mobile applications could be described as particular way in which user utilize the mobile application. Informational, functional and temporal dimensions were the measurements used to identify the usage pattern.

It was understood from the table 01 that overwhelming majority (86.67%) of the users gained information about availability of uzhavan application through government extension officials. Subsequently 08.89 per cent of the users with the help of social media became aware of the presence of uzhavan application and 05.54 per cent of the users through friends/ neighbors came to know about the existence of uzhavan application. From the Table 01, it is reported that majority (60.00%) of the users were not aware of any other mobile agricultural applications than uzhavan app, 40 per cent of users were conscious about availability of other m-agri apps available in various platforms like Google play store, Apple store etc. From table 01, more than half (53.33%) of the users had

Informational dimension:

Table 01: Distribution of uzhavan app users according to their temporal dimension:(n= 90)

S.No.	Informational dimension	Category	Number	Per cent
1.	Source of information of availability of uzhavan app	Government Extension officials	78	86.67
		Private Extension staffs	00	00.00
		NGOs	00	00.00
		KVK	00	00.00
		Neighbors/Friends	05	05.54
		Fellow farmers	00	00.00
		Social Media	08	08.89
2.	Awareness towards other agricultural applications	Aware	36	40.00
		Not aware	54	60.00
3.	Source of agricultural information before using uzhavan app.	Government Extension Officials	48	53.33
		Private Extension staff	04	04.45
		NGO	00	00.00
		KVK	00	00.00
		Neighbors/Friends	17	18.89
		Fellow farmers	11	12.22
		Social Media	10	11.11

Functional dimension:**Table 02: Distribution of uzhavan app users according to their temporal dimension:(n= 90)**

S.No.	Temporal Dimension	Category	Number	Per cent
1.	Preference of language	Tamil	78	86.67
		English	12	13.33
2.	Registration on uzhavan app. for receiving benefits	Registered and benefited	63	70.00
		Registered and not benefited	10	11.11
		Not registered	17	18.87
3.	Frequency of usage of subjects in uzhavan app.	Using 12 subjects	00	00.00
		Using 11 subjects	18	20.00
		Using 10 subjects	03	03.33
		Using 09 subjects	04	04.44
		Using 08 subjects	15	16.67
		Using 07 subjects	20	22.22
		Using 06 subjects	13	14.44
		Using 05 subjects	03	03.33
		Using 04 subjects	10	11.11
		Using 03 subjects	01	01.11
		Using 02 subjects	03	03.33
		Using 01 subject	00	00.00
4.	Most often sought subjects present in uzhavan app	Subsidy scheme	51	56.67
		Benefit registration	10	11.11
		Fertilizer stock	06	06.67
		Market price	05	05.56
		Crop insurance	04	04.44
		Agriculture news	04	04.45
		Weather forecast	03	03.33
		Seed stock position	03	03.33
		Customer hiring center	02	02.22
		Asst. Agri/ Asst. Horti visit	01	01.11
		Reservoir levels	01	01.11
		Feedback	00	00.00

depended on government extension officials as their source to get information about agriculture before the existence of uzhavan app. followed by 18.89 per cent of the users hinge on their neighbors/friends to know about latest agricultural information. About 12.22 and 11.11

per cent of the users had contingent on fellow farmers and social media as their source of agricultural information respectively. Few (04.44%) of the users has used private extension officials as their source of agricultural information before the existence of uzhavan application.

It is evident from the table 02 that, majority (86.67%) of the users were fond of Tamil language and they utilized Tamil for exploiting uzhavan app. Few (13.33%) users reported that they were comfortable with the contents present in the English. It is noticed from the table 02 that majority (70.00%) of the users used the benefit registration section, thereby registered and benefited. The finding shows that these two sections (subsidy scheme and benefit registration) were vitally responsible for the repeated utilization of uzhavan app by the users. About 18.87 percent of the users stated although they were aware of the agricultural schemes and subsidies they did not have the interest to register for receiving benefits. About 11.11 percent of the users had registered but not benefited, this might be due to two valid reasons, inappropriate selection of agricultural schemes or they failed to file the required document which was essential for the scheme. It is understood from the table 00 that 22.22 percent of the users had utilized seven subjects/sections present in the uzhavan app. About 20.00 percent of the users had utmost exploited 11 subjects present in the app. About 16.67 percent users had utilized 8 of the subjects present in the app according to their need. About 14.44% and 11.11 percent of the users make use of six and four sections extant in the uzhavan app, respectively. Very few users (04.44%, 03.33%, 03.33%, 03.33% and 01.11%) utilized nine, ten, five, two and three subject's existed in uzhavan app, respectively. It was noticed that no users have reported that they have completely logged into all the subjects. No users had stopped their utilization with one subject. It is evident from the table 60 that more than half of (56.67%) of the users often pursued subsidy scheme section and 11.11 percent of the

users regularly strived for benefit registration section. Rest of the users contributed for below one-tenth proportion like (06.67, 05.56, 04.45, 04.45, 03.33, 03.33, 02.22, 01.11 and 01.11 per cent) for frequently pursuing the fertilizer stock, market price, crop insurance, agricultural news, weather forecast, seed stock position, customer hiring center, asst.agri/asst.horti visit and reservoir level sections respectively. It was found that no user had sought to browse feedback section. The finding showed that nearly three-fourth (74.44%) of the farmers had utilized the subsidy section, benefit registration and fertilizer stock section to the maximum level because they felt that these sections were more essential and useful to them compared to other sections.

The data for the study was collected during January 2020, and from the table 01, it is evident that majority (83.33%) of the users downloaded the uzhavan application immediately, as soon as they came to know about uzhavan app and its salient features. Most of the users were attracted and got interest instantly towards the uzhavan application. Few users (16.67%) mobilized the uzhavan app after sometimes/due to peer pressure, though they were aware of the existence of uzhavan app they spared some period of time to curb whether uzhavan app is really useful or not. Table 01, indicated that nearly half (51.11) of the uzhavan app users were using the uzhavan application for a period of 7- 12 months, 44.45 percent of the uzhavan app users had experience over 13- 18 months in using uzhavan application and very few (04.44%) of the users were subjected to less than 6 months period of handling the uzhavan application. From the table 01, 37.78 percent of the users had the habit of using uzhavan

Temporal Dimension:**Table 03: Distribution of uzhavan app users according to their temporal dimension:(n= 90)**

S.No.	Temporal Dimension	Category	Number	Per cent
1.	Time of download of uzhavan application	Immediately after knowing about uzhavan app	75	83.33
		After sometime / Due to peer pressure	15	16.67
2.	Duration of usage of uzhavan app.	7-12 months	46	51.11
		13-18 months	40	44.45
		< 6 months	04	04.44
3.	Frequency of usage of uzhavan app.	Often daily	10	11.11
		Daily (once)	30	33.33
		Once in two days	34	37.78
		Weekly	15	16.67
		Fortnightly	01	01.11
4.	Session of usage of uzhavan app.	Morning	37	41.11
		Afternoon	10	11.11
		Evening	30	33.33
		Night	13	14.44
5.	Duration of usage of uzhavan app. per day	<5 minutes	03	03.33
		6-10 minutes	32	35.56
		11-15 minutes	26	28.89
		16-20 minutes	15	16.67
		21-25 minutes	08	08.89
		26-30 minutes	06	06.67

application for two days once, one-third (33.33%) of the uzhavan app users used the uzhavan application daily. About 16.67 percent of the uzhavan app users were found to have the custom of logging into the uzhavan application weekly. About 11.11 percent of users urged to use the uzhavan application often daily. Nearly one per cent (01.11 %) of the users had routinely using the uzhavan application every fortnight. It was exposed from the table 01 that 41.11 per

cent of the users used the uzhavan application during morning, one-third (33.33%) of the uzhavan app users spent their evening for exploiting the uzhavan application subsequently during night. About 14.45 percent of the users had the habit of using Uzhavan application, 11.11 percent of the uzhavan app users made use of their afternoon hours for utilizing the uzhavan application. It is apparent from the table 01 that more than one-third (35.56%) of the users, used

the uzhavan application for 6-10 minutes per day. About 28.89 per cent of the users used to spend 11-15 minutes per day in uzhavan application. The users devoted 16-20 minutes per day for using uzhavan application has contributed to 16.67 per cent. About 08.89 and 06.67 per cent of the users expend the uzhavan application for 21-25 minutes per day and 26-30 minutes per day respectively. Very few users (03.33%) contributed less than five minutes per day towards uzhavan application.

Before and after the introduction of uzhavan application

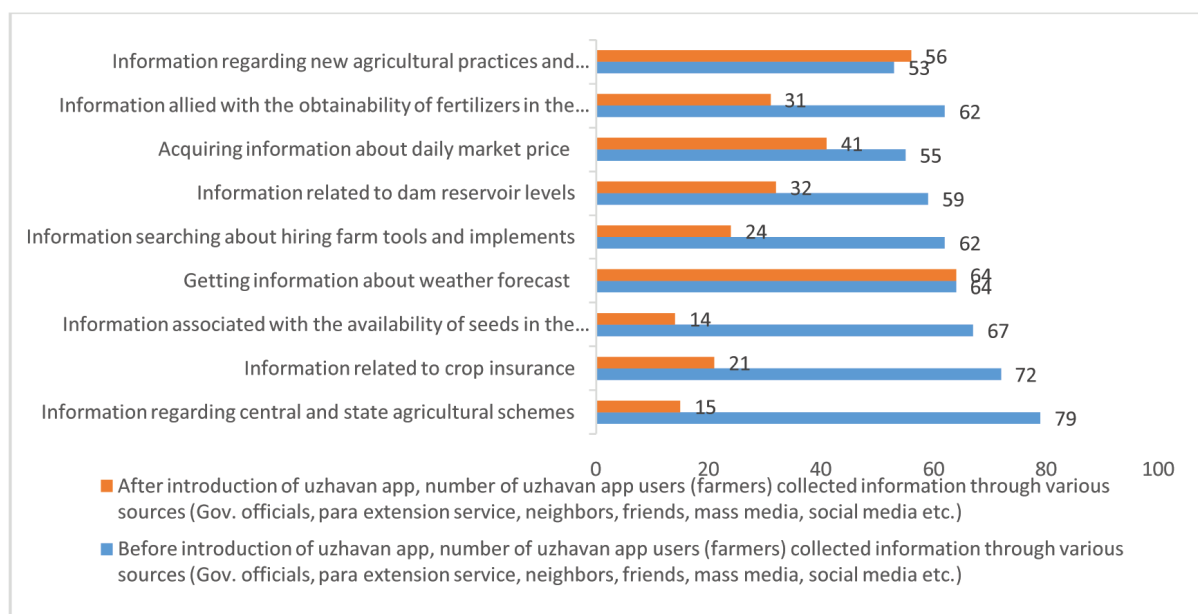
Null Hypothesis (H0): There was no significant difference between users (farmers) collected agriculture related information through various sources (Government officials, para extension workers, neighbors, friends, fellow farmers, mass media, social media etc.) before introduction of uzhavan application and users (farmers) collected agriculture related information through various sources (Government officials, para extension workers, neighbors, friends, fellow

farmers, mass media, social media etc.) after introduction of uzhavan app.

t-test -3.890* St.deviation – 23.56 St. error- 7.85

Alternate Hypothesis (H1): There was significant difference between users (farmers) collected agriculture related information through various sources (Government officials, para extension workers, neighbors, friends, fellow farmers, mass media, social media etc.) before introduction of uzhavan application and users (farmers) collected agriculture related information through various sources (Government officials, para extension workers, neighbors, friends, fellow farmers, mass media, social media etc.) after introduction of uzhavan app.

From the result of depended sample t- test, null hypothesis was rejected and alternate hypothesis was accepted i.e., there was a significant difference between users (farmers) collected agriculture related information through various sources (Government officials, para extension workers, neighbors, friends, fellow



farmers, mass media, social media etc.) before introduction of uzhavan application and users (farmers) collected agriculture related information through various sources (Government officials, para extension workers, neighbors, friends, fellow farmers, mass media, social media etc.) after introduction of uzhavan app. Before the introduction of uzhavan app, farmers have to depend on various sources like Government extension officials, para extension workers, neighbors, friends, fellow farmers, mass media, social media etc., in the search for agriculture and its allied information. This caused more information seeking time. But, uzhavan app was considered to be a consolidated mobile agricultural application where farmers can search agricultural information related to central and state agricultural schemes, crop insurance, availability of seeds and fertilizers in their region, hiring farm tools and implements, daily market prices, weather forecast and reservoir details within a single click. The reduction in number of farmers who collected agriculture related information through various sources after the introduction of uzhavan app indicated that most of the uzhavan app users (farmers) have been convinced by the agricultural information provided through the uzhavan application expect in weather forecast and new agricultural practices and technologies provided in agricultural news section. Hence, the dependency of the farmers with others for seeking information has been greatly reduced with the help of uzhavan application. This result of the paired t-test also coincides with the farmer's perception. Uzhavan application not only reduced the information searching time but also deliberately made awareness about presence of various agricultural schemes (central and state)

for marginal, small, medium and big farmers. Uzhavan application has eased the process of registering and availing agricultural subsidy and benefits for the farmers in Tamil Nadu.

CONCLUSION

Farmers need accurate, brief, timely and trustworthy information to practice and perform agricultural activities. Regional language was more comfortable and convenient for the users because with the help of their regional language, users could exploit every content present in the application to the maximum level. Though majority of the users in the study have completed colligate education they prefer regional language (Tamil) to English, this indicates that presence of contents in regional language would be a deliberate advantage to reach more people. Some of the users had good experience with other m-agri apps like nithravivasayam, m-kisan app, paddy expert system app, cattle expert system app, coconut expert system app etc. Some users who shared that social media especially YouTube and Facebook was their main source of awareness, which made them conscious about the existence of other m-agri apps. The finding of the study showed in a big picture, majority (66.67%) of the users have exploited utmost seven subjects present in the uzhavan app and one-third (33.33%) of the farmers utilized the uzhavan app. In the morning, before going to the field the farmers just log in to the uzhavan app and check the weather forecast, reservoir details, agricultural news, and daily market price, start the day with more updated knowledge. These sections present in the uzhavan app boosted the farmers to look after it during morning hours. This mobile application was a very good initiative taken in the field of

agriculture, the content present in this app was so strong and delivered the vital extension service to the farmers so the positive attributes of this uzhavan app technology could be widely used in this state.

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